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AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 307 — Past Questions & Materials

by Renance

Course: CSC 307

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — CSC 307

Level 300 • First Semester • PastQuestion

i.

AMBROSE ALLI UNIVERSITY, EKPOMA

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

2020/2021 ACADEMIC SESSION FIRST SEMESTER DEGREE EXAMINATION

Course Code: CSC 307 Compiler Construction Time allowed: 2 hours

Instruction: Answer question 1 and 3 other questions.

Given the definition of a sentence using Backus Naur Form as:

Sentence $\rightarrow$ spvp ow. 3.1

where sp = sentence phrase, vp = verb phrase, op = object phrase and sp, vp are

further defined as follows;

sp $\rightarrow$ _ article noun 3.2

vp $\rightarrow$ ~> _verbop 3.3

op $\rightarrow$ article noun 3.4

verb $\rightarrow$ has 3.5

article $\rightarrow$ a 3.6

article $\rightarrow$ the 3.7

noun $\rightarrow$ woman 3.8

noun $\rightarrow$ shoe 3.9

1(a). Use the definition described above to derive a string "The woman has a shoe" (10 marks). |

(b). Illustrate a production with 2 possible of syntactic element using the same definition. (2.5 marks). |

(c) Draw a syntax tree for the derived string in 1(a) (5 marks).

2(a). Consider the following FORTRAN program segment.

REAL NX, JX, COUNT

READ (30,*) NX, JX

COUNT = NX + JX **2.3

IF (COUNT .LT. 2.3) THEN COUNT = NX

Form your terminal table and show the content of the literal, symbol and uniform symbol tables after a lexical analyser has gone through the program (10 marks).

age

indi iii tion of a

i i tokens (iii) Formal defini

(b). Explain the following terms (i) Alphabet (ii) Finding (language. (7.5 marks).

3 (2). With the aid of a detail diagram, elucidate the various phases of a compiler (12 marks).

(b). How is a compiler ported from one computer to another? (5.5 marks).

4(a). Explain the production with the aid of an ex

pression and draw the operator precedence

table (10 marks).

expression $M + J * N$

(b). Write @ two ~column stack algorithm for the'

5 (a). Explain error detection and recovery (5.5 marks),

(7.5 marks).

(b). Write short notes on the following (i) a simple-code generator (ii) storage for temporary

Variables (iii) Storage allocation (12 marks),

6 (a). Discuss local optimization (6.5 Marks). i

(b). Write short notes on (i) Syntactic erro

rs (n(s

(12 Marks),

yntactic error recovery (iii) Lexical errors

wy

4a. With the help of a diagram, describe the various parts of compiler and show how they relate o
one

another. 8mrks

b. What is the difference between a translator and an interpreter? 4mrks

¢. State the functions of (i) linker and (ii) loader in a compiler. 4rarks

5a Convert the following statement to quadriples.

rsam-q/p+txw Smrks

b. Given the produciions below, show how a bottom-up parsing will parse the expression. 6mrks

$a + b \times (c + d)$

$Pp \rightarrow (E)$

$P \rightarrow + \gg \text{identifier } 7 .$

Ee

= Pest

$^{\text{TM}} > Pp$

TP .

_c. explain the difference between loop unrolling and loop jamming . 5mrks

" 6a. suppose a certsin language is made up of the following productions.

$S \rightarrow AX$

$A \rightarrow Vv$

$A \rightarrow vite 8$

$B \rightarrow Ww ' .$

- (i) Waat is the alphabet of this grammar? 2mrks

" (ii) Woaat are the terminal symbois and man termina! symbols. 2mrks

(iii) Define a sentence and hence define a grammar. 2mrks

(iv). Oefine alanguage. 2mrks

oe

b. State and give example each of understanding of:

1, Syntacticerror 2 * mrks

i. Tedeal erro? and 2" mrks

III, Sernaticerror = 2*mrks

i

ate ts purpos®: (S marks)

" P Definca Tee Nanded ddanice, SI :

. . An fige™ s ,

(4 marks)

" } Shile the sigraficant aspects of Reuter Langage

~ 4 What iy Tokonizati Sa 3 y

% atay Tokonnmadon', . catitfenehts OC (3 marks

SA Optimiser atterapts to anprowe he tive and Space. POgMIFEITE nis of 4 program. Mention four oh et

at ewig oe ENE cet 4 marks) fis

sy oe the detaition of Carrere Pree Gramanae (C oo Pisctid ; j (2 marks)

oa (Fhich ef eK ng, C4 HE source errors would be detected at compile time, « . %

a ¥ hich of ae Folk CK SOUECE i . p ime, and which would be

'ted sy oun ime. t

ay P(X) then A © oes ESon , ; ges oye

" : "SSE Sa ; ye * a

else A= X:

(iii) if (A270) X + A: ase oon

; else if (A+ OPN 5 103 f "88 ayy?

else NNN Sieg. Se : Es :

(iv) while ((p- -inlo-9) KE (p!=0)) ake eS xX

p= p-rnenb- ' Beste yo ; ;

/7* assume p points to a struct ots \ Soe ae

2? N70) - et = rs aa : Aaa ¥ \

named node wilh Sot eo 7

/ these fold definitions; eae ys : ere

3 ' vgs aie? Boek' "3 ue f '3

ofl int info: NG See Ot . ii ge \ x

ao ge node * next;*/ ee Soy ASS ae aS aie (6 marks)

t . ; \

i
 ‘: i}
 i ‘ ‘ 1
 Las | Re Nas |
 t ye 4
 t ¥ ‘ I
 | : p Roy ae
 i} ete “
 : sompilee and a preprocessor, |
 1D Wie Lining etenence be nd be roleetell at compile 4
 Y Which of the Following CZF sou sa
 detected at ran time? 4 ; ies }
 (GYR PQs dy : |
 (i) UY <9) then D7 iy Prat wate ae
 else Bos Vy
 Wie a 2 qitvirageay x uaa aE ; at
 if © else (Qe) Xe HE |
 i else.Xo? XQ; 4 4 . |
 } sCiv) for ((p->news>0) or (p!=0)) | rena ae
 ie : PD > pesnenks is See ee “4 oles
 d fia! HR Wyn in modern
 , Ae finiqiae perfomidice does the first phase. of a compiler frontend play in m |
 processing? (Sinks)
 b)-Consider the language $E \rightarrow E+E \mid e$ ks)
 derivation for the sentence $(id*id)+id$. (7MKS). Fete oe
 r ime : Ss |
 Write a regular expression for a language of Soe from 0 to 199..3mks)
 * £|(E)| id. Write the lcfimost derivation and rightmost
 6. a) Whiy do we need syntax trees during compilers construction? (3mks)
 by Describe in detail the group phases during compiler construction. (6mks) A “aad
 c): Construct the NFA for the regular expression $011(0 \mid 1)^*$ using Thompson methoy and conve
 it into'minimized DPA, (6mks) oP ie
 j ; AMBROs
 } — DEPAR MENT LUUNIVERSITY EKPOMA
 j 201672017 SESSION ae: OF COMPUTER SCIENCE |
 TSEMEsT ER DEGREE EXAMINATION
 wie cs
 C307: COMPILER CONSTRUCTION |

Instruction: Attempt Questi

peg 1001 Any Other 3 Questions 'Time Allowed: 2hrs

i a) State the english description of the sets of sttin juiced by the regular e
ce x

i) (aiip}* ex(nulp) ii) e(cld)*c. (4imks) eet 's aaa

es) Eoltin w hya Courier is OMine Whereas an Interpreter is Online, (6mks) |

OWeseribe how compiler construction vinbodies the theory, practice and branches of Computer
Science (6mks)

What de you'consider when genelating codes (ura specific rglocatable machine code? (3mks)

e) Create a Lexeme, token and pattern for the followings !

progran} xmiult (input, output); var "integer; begin L "L*(45/MULTL); end. (5mks) |

f) Write an algorithm for a typical left-factoring, (Smks) : i

g) Convert the regular expression (era) nam i into NFA, 'Then convert the respective NBA to DFA
agd minimize, 6mks pele Shane |

st

1

2. a) Show the compiler output and interpreter output for the following source code:

i while (CSD)

Tie oo pears :

Seout<<C <D<<end!: pe

b} Suppose the grammar G:S 2 SS+]8 SH A a and the ing: See a 4.

3} Stare pe telanicen? derivation for the string . it) Sine a paatse tee Tor the string. (Sinks)

c) Explain why every production rule ofa Context-lree-Grammar is of the form $V \rightarrow w$. '(3mks)

Se Unaksy 2 7

3. a) List and explain the steatchics to redeem errors during compilation. (4mks)

b) Draw a DPA for the regular expressioivi) $k(k|h)^*k$. Gimks)

q) Draw an NFA for the regular expression fi) $((x y)^*z (m | n) (a |b))^*e$. Gmks)

c) What are expected of a standard podels compiler? (oinks)

\$. a) List the elass(es) of grammar that this grammar can exist: ce a \$: ala. mks)

Why i is finite automata relevant to compiler construction? (4mks)

'oy each of the following language rules, identify the phase of the compiler that should contin that a
ram conforms to that rule and why that part of the compiler is the best place for that check: |

-4 function is called with the correct number of arguments. \

yUnderstore characters (-) may appear in the middle of identificrs, but not at the boginning orjend,

Ulivery variable niust be declared before it is used in the program (the classic C or Pascal rate).

DxiAsigipon! statements must end with a semicolon (;).(2mks each), \

ta

Ming ; 4 |

PACU OY ALLE UNEYERSEPY Pe Nome,

1 Dep : Wen A. |
 ROWSE Ste TR UMENT gpATURAB SCIENCES ties .
 ONESSTON Ua reee Shay OF COMPUTER SCIENGCH _¢
 7 , oe Th) ard ow ¢
 ei UC ag] ES EMCUNDERGRADUA TE EXAMIN ade
 Instruction: Answoy J eestion Disa "Mier Constuetion atk |
 SEY Wifey TITHE Onestions Fine Attioweeds > 4
 ia} : tint iy the Uionenee between an ISSN Se aN ir
 hy Show afinite Slat Mehine yeahs slate bs bal acomptler? bimikes
 cach vase thoinput alphabet is 101) 4) Stings i oF table i = Kr cach of the fubtowings fangs fin
 ea and Auiitoe af ace ris ait ves 'tutte he Wt UN Hee Connerntive OMIGS. tt) Stren 1 COMME ey re i
 Mey
 Yc) Ad iitetpreter is ontine, Diseusy. 0 (3 marks eset)
 ' § " *)
 VY dp Discuss the wens iin vompiler comttuetion if ark ')
 YS ' \ 0 ee se ah (5 mark ')
 «) Show a mbly Herriterten Hepes WaMON Loe a vehine of Your chwice, corresponding to the
 following CACO) statement: lor (yp iy ye Jey : '
 he Co nirks)
 W741) Create a texeme, token and patter for the Following
 Pregecda psub Capatcoutpaby vat xin begin \ NUSUIUN-9): crn
 ye) Show the token classes, of words), PALOUÉ DY the leyieal analysis
 source inputs if (Add = Ted) 7" Add | 16 47s,
 (5 nuarigs)
 phase corresponding: to thins C54 4
 (9 marks)
 day Describe any three {ypes of errand ponenited in compilation process,
 Xb) Ifa portion of the input toa Cit chmpiter looked tike this: Eo dh it G:
 Show fhe outputola CAC OC compiled in aty typical assembly language , (4 marks)
 ye 0) Which Hinde state mitehing is call | Deterministic. Why isi referred tas such'? 5 (Snark)
 (6 nrivks}
 i:
 Aa) Explain the teclaiaacy used in doep aptinization, i | (5 marks)
 hy Write short notes (with descriptions) on "Token, Pattertis, and Lexenics. i(G inarks)
 c) Give parse tree for the string S242 4S Se ; he erie eee '(4 marks}
 V4) Argue that the hruguage ofall stei ys of the form {bef ped} (equal number of '{" aid") is not stat
 t. :
 regular, Give CPG that accepts precisely that languayse, (Gmarksy ule!
 b) i) Write a rept expression for rll strings oF as and bs which contains the substring abba -f

ii) Write a regular expression for all strings of x's and y's where every y is immediately followed by at least one x.

least 4 x. \?

sii) Write a regular expression for all strings of p's and q's which contains an odd number of y's. R
Saree TE take the one who saree (3 marks each) *

c) The output from code generation is usually assembly language or relocatable machine code. What do you think?

you consider when generating code for a specific machine? (4 marks) (4)

i} site tak Ge a | Fy

i ' '

&) \$a) Genefute ianuaye Jor the following pram: Ge ' ody

ae, Slee OSE + [null " S->bAc eae ie Ma

\$2-> 152.0] null Avge ey ot) GO marks each)

yy View are he Hanetions per farmed by the analysis phase? i tos (5 iawks)

cp isrigtly discuss what the potential dud vantiyges/tisaclvictlagen are OF bollom-up versus suplowy

Pane' pENELALOE\$, wen ' miuks)

9 Ga) Detling amy aunatalead purse trey, Give an bade nit ae ae (3 marks)

oh) A pineal poipilation inn phliwe " el a , anne _ Whig belore (he Next phage beyinns (riher
 "than being, iticayaye,t yr tosacal] Vill subsequent phases), Why anight a phase (or ser of phases)
 be

guplenented ja pases palin? ' uf (} narks)

| H

4 ne

} MN

©) Starg the Uefininy: deference banween a compiler and a preprocessor, "e Mh Marks)

Q) Which of the following C/C++ errors would be detected at compile time, and which would not?

letaeuyt Oru tine mes rity cba \

Wa Maas, ¥

2° MDT) honey f

: : DIST me Yas 4 : : , : '

6 AY Q-OPN oe :

: Yoon Y a . , ' '

ele MO OVX © a: A heh

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j eelse Nt N/A): a :

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ot ON) for (fps news) or (pl 0)) , os i

be pe nent: ay nee sé (6 tate)

es ¥

P'

AM : '

FACILE UNIVERSITY, KKPOMA

DReART Y OF PHYSICAL SCIENCES

aie MENT.OF COMPUTER SCIENCE

/2019 FIRST SEMESTER EXAMINATION

exc AWE

Inserncs Be aaa ge and Lane ciated ite: 30/05,

E E and any other, TUREE amestions Vime: Ze

- Ha. Desie the following terms (i) Com i; ay (ob vtiaed i

piller (ii) Assembler (iti) fnterpreter (iY) Regular Gramimet (4 iharks) ;

Ub. During the compilation process, operator precedence parsitig is always a tedious task to be exdcuted by the compiler Performa

Design sy algorithm that cart help this process in-any compiler desigi. G Mar))

ce. The Mission Code (MC) is a product of one of the phases of compilation. You are required to!

ei Dae this phase (1 mark) Fee je £ é i a

scribe its relationship with other phases (2 marks) ze "ach

(iii) Practically generate the jnission code for $y = xth * z$ sey mys ks}es ce

_ 2. Given that yeb™ aac. You are required to: " ee tq g.

(i) Geacraie tts appropriate grammar a mark)*. Masti ee WG ate ',

{ii) Practivally show if the grammar) is. ambiguous (2, anaes et ees

Gil) Briefly explain how ambiguity can be checked and remove from any erainirnar Q ark ik-m

we Se. Asa ponae engineer im 'Mikeson Technologies. "you were given that a compiler was: — wy a grant ce G eed each

ired to: \ } a oe

id -was giv en to it. You ace. e requiring

Ett /® *c7id anda language of idtid*

i) Ger cate the leftmost derivation of the language (

Gi) Gersrate the "eae? cape of the lahat, Se That

matks

Zs Gs Sarde

{Esa +

Gv} Using the Yast' te di

marks)

{. With a strong arguitent, Row w

whether the' gramuiar S-oxSSee js ambiguous with the language. 8! OSes

{9 java compiler. (3 marks)

wh, 4 Souk:

seed for the major operation in # pr

/ da, Using the Jast five digit of your matriculation number as a

z alla

he lexical analyzer 'on the written panera: (SKE

; a

choice and explain the function of ϕ

program of your

b, After the lexical analyzer has processed the pega written in 2a. show

{i) Terni,inal 'Table (2 marks) : Lost es ee

fii) Literal Table (2 marks) 3 eae Sis

Gif) Une Zorm symbol Table (2 marks) ; R saree ALIN : : ate

ame is a-valid identifier in the write propeam iy 2a. Design is slate aed We. shew ihe

c. Assuring {hat your surn ; ji

pee for each state. (4 marks) ace

3a. Deserjhe the job of syntax analy?

"b. Consider the production given below.

zex and stute its (wo major function on the program Written ie 2a, Vamahs)

1, F $\rightarrow$ » (6)

2 $\rightarrow$ (identifier) ; eS

Q

2

my

" hor '

ee Se ad \

bp ca : \

5.T RK

6. Top TE

7 E> LY d

8b > om ive digit oF

jsuming % is the last five digit «

a Using botiom Up parsing method illustrate how' the expression $XFY\yen*/$ can be parsed, assuming X '5 the lest five dig :

your matriculation number, $\yen=2001$) and Z10, (5 marks) tne 4 viatles)

re ©. Practically differentiate between the parse tree and the syatax free Mec enth beipenorate (ror Gustine'. 2.

4a. Bri 'My describe semantic analyzer and explain why some revel

declared is used ina Program but

ssemantic analyzer flag errors if vatable that is not

allow the addition of Wo.uperitid of differentdatatype. (4 marks)

b. Using the production in question 3b, show the content'of semantic stack of $a^*(bc)$ (4 marks),

~€- Consider the following Cit program ; ;

Public Static void Main (String[] args): >.

{i

Double B, C, Maric, PhoneNumber 4 "eh %
 Console. WriteLine ("Input the last five digit of your intiio number*
 Matric=Console.Convert. Ti odouble(Console. Readlino());: vi {
 C=100;
 PhoneNumber=Matric+C*B;
 Console. WriteLine ('My Phone Numbe: :
 Console. WriteLine ("Congratulation for g
 - PhoneNumber) Bo
 ating your phone nun
 ; erenitiate: bet f ' i 8 & compiler 'and:
 explain the tool used in rearranging inte : i % é
 'amin 4e° and
 soniment on their optimality. (5 marks
 . As a software engineer jn Hawei Techno gies, 'you
 ist five digit of your matriculation 'number, Desi
 { cvlerate the stack output. (5 marks)
 \$*X Wher X isthe
 OSTlix- mechanism an.*
 6. With: a clear grammar definition, practically show the result
 _ » Hrivtly describe when a semantic analyzer is said to
 us -ç jn Compiling the program in 4c. (4 marks). ; he i
 jf \sa Software developer in Neptune Software Company
 ai we and matriculation nymber.
 g rithns. (5 marks)
 5
 é pany, YOu were givan that a compiler is to Walldats ONT your
 Design a state machine that will ehable (his Compiler to do its ib "ana shows Hd state
 >. 0 ie ee i be ag
 se ° Y
 Bet idan ee sa TACULROSE ALLI UNIVERSITY |
 Go AR THEY OF Ratural SCIENCES «| a,
 Ps gotannots session ree ENT OF COMPUTER SCIENCE | 2
 ç | COCSEMBSTER t UNDERGRADUATE EXAMIN Aten
 ustrcton Answer Question ! fion I 307; Complict' Construction !
 r "nS tny other THREE Questlans | Time Allowed: 1
 day Describe at minpiler With regards te Us ; | eae
 : : fee and programurinys languages, ¥ ,
 Deseribe thie phasey oF Wedinp cr inter Lop.) 3 niarks)
 iB Binal then Pier in the' analysis/aynthesis model of compilatito, With the aid of ay

¥ ddA compiler is oMing, Discuss, vo : | (4 pei

. | wks

5 dy Staic why a Lenical' Analyzer is kept as an independent module,» wo marks >

+7e p) Show an avembly language representation. fora machine of your choice, corresponc: rn Ho the)

following C/OH statement' while (XSitHD) = 2X Wy i ct

(6 marks) ay Sate St oe

Hy A Crate alexemo, token and paitern for the loliowing:.. uf d

| Progra psn (input, 'oulpul)s, var ysreal: bein Y= YASUMY* | 1); end, 2 muarks)

5) Show the token classes, or "Words", put out by the lexical Ela phase Sr lo this C+

source e inputs if (Sum: BALE is Sun=: 133) ! - | (5 marks)

+ 28) Tostraie with tieand of b dingrax the difference b 1 compi and an an int inser (6 mks)

- -=}; sag ~b) lfa portion of the input toa Chee ct J ag

Stiow. the output ofa CIC 'compiler, in d marks).

- ae ¢) Deseribe a ivratal Renee Ao mats)

et

. Be 3a) Briefly Aeatribaihe ey dev lopm

ate a.) Show the compiler outputit and ff

* for G=13j<=6; uae

| SS ¢) ee ine'the following i} Linker eek :

| at Ey 4 € '

Seca) Argiic, that the jangtigs of alt WINES, of the form-{ {...4}.. 3) (c: a number of or'(and' } is nol \

pt regular. Give CFG that uveepts precisely that langtages . ' Aston Cae ea te oxurks) i

j pe | b) Gorsises the Three-Address! Cate TAD eh : ae y |

| | og fd: tompis=y nts ue :

i temp3 = einp li: a

4 pad "¢, if temp]. = temp3 voto 200 Ectee 1

' oo Ap A temp2 2 tenip) * x3 ep cai |

: : by) 32. goto 300 EO AN, }

i beh oe) 200temp2 #2 ; | |

Ue ', 300 temp4 := (emp2 + temp3 att aes é ' i

Ey What would ¢ the outcome of ti pemoving gle redundant item temp) ci) removing the |

"redundant code? ?

(S marks caci)

5 chi

pleat machine « ate, What do |

= cy The output from code: genemtion is ustk aly assehibly language'

you coh Fell chal pie is a specific machine? a) re | (A marks)

Su) ei the ps ou a th w interp utput for the following Crt sotinee ents n

cout aN << B<s cud) ts yak i |

; i | sea' '4

i